



Master thesis

Master of Science (M.Sc.)
Department of XYZ
Major: ABC

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with related
text

Topic:

Title Line 1

Title Line 2

Author: Author Name

Matriculation Number: 123456789

First supervisor: Prof. Dr. Alpha Beta Gamme

Second supervisor: Prof. Dr. Alpha Beta Gamma

Submitted on: xx.xx.202x

General Instructions:

1. The total thesis document should have at least 100,000 characters excluding references and appendix (if any).
2. You must have exactly the mentioned sections with the mentioned number of paragraphs.
3. Observe the report page limits (minimum 60 pages without citations/references and maximum of 140 pages excluding citations/references).
4. In case of figures, keep your raw data table also stored as excel file in this repository.
5. Each paragraph must consist of 7-10 sentences.
6. Add references where required.
7. Ensure that the total AI content is strictly less than 20%.
8. Total references should be between 80 and 150.
9. Use latest references with 80% references later than 2020.
10. Comment this Introduction section before making the final submission in the-sis.tex file.

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Berlin, xx.xx.202x

.....

(*Signature*)

[your name]

Abstract

Four sentences of background and importance of this topic.

Four sentences defining the gap analysis (work not done in field) and your problem statement.

Four sentences about the methodology of your work.

Six sentences of Results and significance of your results.

Two sentence summarizing your contribution.

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write a one page abstract
- Approximately 20 sentences with importance, background, current state, gap analysis, problem statement, your solution, your methods, your results, and improvement in field being discussed.

Zusammenfassung

Da die meisten Leuten an der TU deutsch als Muttersprache haben, empfiehlt es sich, das Abstract zusätzlich auch in deutsch zu schreiben. Man kann es auch nur auf deutsch schreiben und anschließend einem Englisch-Muttersprachler zur Übersetzung geben.

Contents

List of Figures	xiii
List of Tables	xv
1 Introduction	1
1.1 Motivation	1
1.2 Objective	1
1.3 Thesis Scope	2
1.4 Purpose and goal	2
1.5 Outline	3
2 Theoretical Background	5
2.1 Background	5
2.1.1 Definitions	5
2.1.2 Important subheading 1	5
2.1.3 Important subheading 2	5
2.1.4 Important subheading n	6
2.2 Existing Methods	6
2.3 Evaluation Metrics	6
2.3.1 Subsection 1	6
2.3.2 Subsection 2	6
2.3.3 Subsection 3	6
3 Related work	7
3.1 Section 1	7
3.2 Section 2	7
3.3 Section n	7
4 Research question	9
4.1 Problem statement	9
4.2 Research question	9
5 Methods	11
5.1 Research methods	11
5.2 Datasets	11
5.3 Data Preprocessing steps (if any)	11
5.4 Experimental Settings and software used	11
5.5 Network Architecture (for deep learning models only)	11
5.6 Other details for reproducing your methodology or results	11
5.7 Techniques for Data Analysis/Visualization	11
5.8 Evaluation Metrics	11

6	Experiments and results	13
6.1	Model Training	13
6.1.1	Environment	13
6.1.2	Training process	13
6.2	Experiments	13
6.2.1	Subsection 1	13
6.2.2	Subsection 2	13
6.3	Model Generalization	13
6.4	Comparison with Existing Techniques	13
7	Conclusion	15
7.1	Summary	15
7.2	Dissemination	15
7.3	Problems Encountered	15
7.4	Outlook	15
	Bibliography	19
	Annex	21

List of Figures

1.1	Component X	2
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List of Tables

1 Introduction

One-Two paragraph(s) introducing the purpose and theme of the research.

1.1 Motivation

Two-Four paragraph(s).

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1.2 Objective

One-Two paragraph(s).

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'Figure 1.1 shows Component X as part of ...'

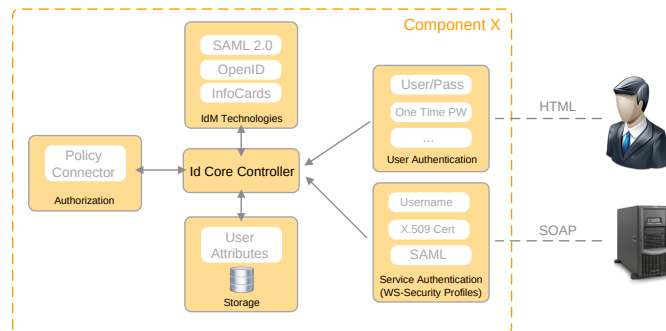


Figure 1.1: Component X

1.3 Thesis Scope

One-Two paragraph(s).

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1.4 Purpose and goal

One-Two paragraph(s).

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1.5 Outline

The 'structure' or 'outline' section gives a brief introduction into the main chapters of your work. Write 2-5 lines about each chapter. Usually diploma thesis are separated into 6-8 main chapters.

This example thesis is separated into 7 chapters.

Chapter 2 is usually termed 'Related Work', 'State of the Art' or 'Fundamentals'. Here you will describe relevant technologies and standards related to your topic. What did other scientists propose regarding your topic? This chapter makes about 20-30 percent of the complete thesis.

Chapter 3 analyzes the requirements for your component. This chapter will have 5-10 pages.

Chapter 4 is usually termed 'Concept', 'Design' or 'Model'. Here you describe your approach, give a high-level description to the architectural structure and to the single components that your solution consists of. Use structured images and UML diagrams for explanation. This chapter will have a volume of 20-30 percent of your thesis.

Chapter 5 describes the implementation part of your work. Don't explain every code detail but emphasize important aspects of your implementation. This chapter will have a volume of 15-20 percent of your thesis.

Chapter 6 is usually termed 'Evaluation' or 'Validation'. How did you test it? In which environment? How does it scale? Measurements, tests, screenshots. This chapter will have a volume of 10-15 percent of your thesis.

Chapter 7 summarizes the thesis, describes the problems that occurred and gives an outlook about future work. Should have about 4-6 pages.

2 Theoretical Background

This chapter provides the reader the background knowledge to understand technical concepts applied in the thesis.

2.1 Background

2.1.1 Definitions

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2.1.2 Important subheading 1

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2.1.4 Important subheading n

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2.2 Existing Methods

At least 8-10 different cited papers for techniques you deploy for problem solution with 6-8 published after 2019.

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2.3 Evaluation Metrics

2.3.1 Subsection 1

2.3.2 Subsection 2

2.3.3 Subsection 3

3 Related work

In this chapter, we discuss the relevant literature review for the chosen topic.

Identify the important subcomponents (e.g., in terms of methods, evaluation, preprocessing or visualization) of your thesis and for each subcomponent, make a separate section with subsections as shown below.

The summary of the literature is shown in Table ??.

3.1 Section 1

At least 8-10 different cited papers for techniques you deploy for problem solution with 6-8 published after 2023.

Group in terms of similarity in approach.

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3.2 Section 2

At least 8-10 different cited papers for techniques you deploy for problem solution with 6-8 published after 2019.

Group in terms of similarity in approach.

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3.3 Section n

At least 8-10 different cited papers for techniques you deploy for problem solution with 6-8 published after 2019.

Group in terms of similarity in approach.

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vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetur odio sem sed wisi.

4 Research question

This chapter introduces the research question ...

4.1 Problem statement

4.2 Research question

5 Methods

5.1 Research methods

5.2 Datasets

5.3 Data Preprocessing steps (if any)

5.4 Experimental Settings and software used

5.5 Network Architecture (for deep learning models only)

5.6 Other details for reproducing your methodology or results

5.7 Techniques for Data Analysis/Visualization

5.8 Evaluation Metrics

6 Experiments and results

6.1 Model Training

6.1.1 Environment

6.1.2 Training process

6.2 Experiments

6.2.1 Subsection 1

6.2.2 Subsection 2

6.3 Model Generalization

6.4 Comparison with Existing Techniques

7 Conclusion

The final chapter summarizes the thesis. The first subsection outlines the main ideas behind Component X and recapitulates the work steps. Issues that remained unsolved are then described. Finally the potential of the proposed solution and future work is surveyed in an outlook.

7.1 Summary

Explain what you did during the last 6 month on 2 or 3 pages!

The work done can be summarized into the following work steps

- Analysis of available technologies
- Selection of 3 relevant services for implementation
- Design and implementation of X on Windows
- Design and implementation of X on mobile devices
- Documentation based on X
- Evaluation of the proposed solution

7.2 Dissemination

Who uses your component or who will use it? Industry projects, EU projects, open source...? Is it integrated into a larger environment? Did you publish any papers?

7.3 Problems Encountered

Summarize the main problems. How did you solve them? Why didn't you solve them?

7.4 Outlook

Future work will enhance Component X with new services and features that can be used ...

Bibliography

Annex

```
<?xml version="1.0" encoding="UTF-8"?>
<widget>
  <debug>off</debug>
  <window name="myWindow" title="Hello Widget" visible="true">
    <height>120</height>
    <width>320</width>
    <image src="Resources/orangebg.png">
      <name>orangebg</name>
      <hOffset>0</hOffset>
      <vOffset>0</vOffset>
    </image>
    <text>
      <name>myText</name>
      <data>Hello Widget</data>
      <color>#000000</color>
      <size>20</size>
      <vOffset>50</vOffset>
      <hOffset>120</hOffset>
    </text>
  </window>
</widget>
```

Listing 1: Sourcecode Listing

```
INVITE sip:bob@network.org SIP/2.0
Via: SIP/2.0/UDP 100.101.102.103:5060;branch=z9hG4bKmp17a
Max-Forwards: 70
To: Bob <sip:bob@network.org>
From: Alice <sip:alice@ims-network.org>;tag=42
Call-ID: 10@100.101.102.103
CSeq: 1 INVITE
Subject: How are you?
Contact: <sip:xyz@network.org>
Content-Type: application/sdp
Content-Length: 159
v=0
o=alice 2890844526 2890844526 IN IP4 100.101.102.103
s=Phone Call
t=0 0
c=IN IP4 100.101.102.103
m=audio 49170 RTP/AVP 0
a=rtpmap:0 PCMU/8000

SIP/2.0 200 OK
Via: SIP/2.0/UDP proxy.network.org:5060;branch=z9hG4bK83842.1
;received=100.101.102.105
Via: SIP/2.0/UDP 100.101.102.103:5060;branch=z9hG4bKmp17a
To: Bob <sip:bob@network.org>;tag=314159
From: Alice <sip:alice@network.org>;tag=42
Call-ID: 10@100.101.102.103
CSeq: 1 INVITE
Contact: <sip:foo@network.org>
Content-Type: application/sdp
Content-Length: 159
v=0
o=bob 2890844526 2890844526 IN IP4 200.201.202.203
s=Phone Call
c=IN IP4 200.201.202.203
t=0 0
m=audio 49172 RTP/AVP 0
a=rtpmap:0 PCMU/8000
```

Listing 2: SIP request and response packet[?]